

SIA: Selective Image Analogy

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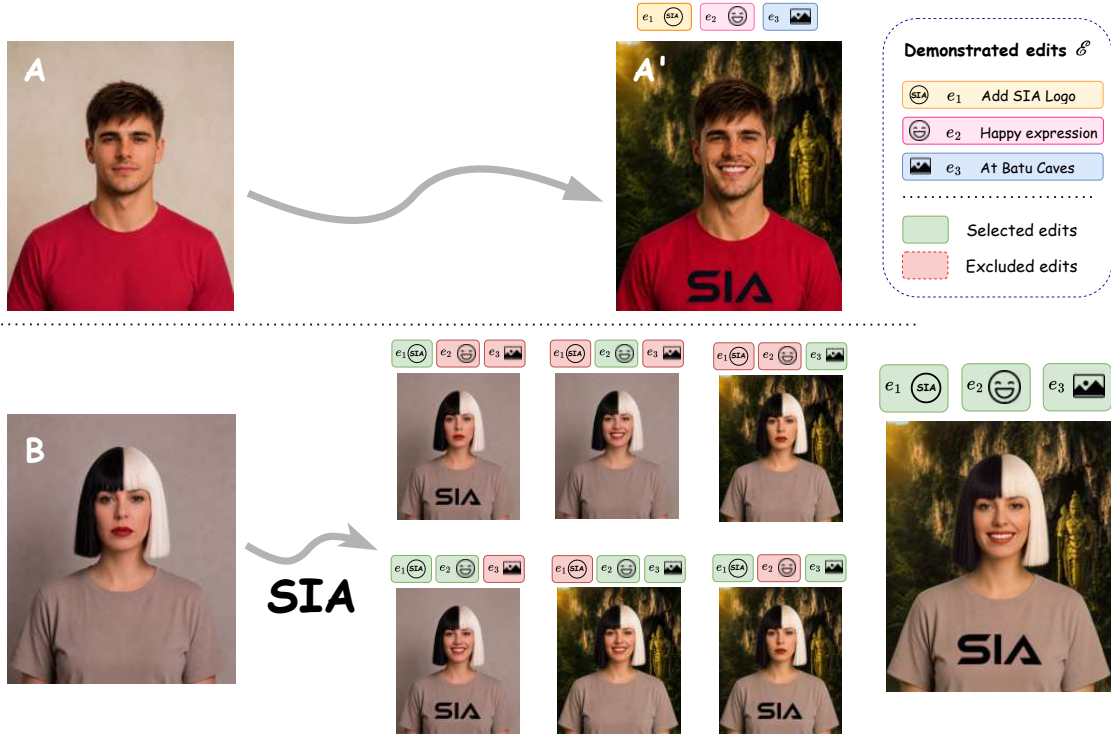


Fig. 1. Teaser of SIA. A single reference pair (A, A') may contain multiple edits at once. In this example, the changes include adding a logo, changing the expression, and replacing the background. Given a new query image B , SIA lets the user choose which edits to transfer. The same demonstration can produce different outputs, ranging from single-edit results to combinations of selected edits, while leaving the remaining edits out.

Image analogy transfers a demonstrated transformation from a before/after pair (A, A') to a new query image B . Existing methods assume all changes in (A, A') should be fully transferred, limiting flexibility when users want only a subset of the demonstrated edits. We introduce *Selective Image Analogy* (SIA), a new setting where a single demonstration pair encodes multiple edits, and the user selects at inference time which edits to apply to B . To support this, we curate a multi-edit analogy dataset with full-edit and partial-edit targets. We then train a model that first learns to transfer the complete demonstrated transformation, and then gains selective control through a mask-aligned suppression mechanism: a token-gated LoRA that

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restricts adapter updates to suppressed edit tokens during denoising, paired with a concept suppression vector that weakens the selected tokens in the text embeddings before denoising. Together, these components allow SIA to suppress unwanted edits that remain visible in A' , while preserving the user-selected ones, all without requiring a new demonstration pair for each edit combination. We perform extensive quantitative and qualitative evaluations along with user study to demonstrate the efficacy of our approach.

1 Introduction

Image editing plays a crucial role across a wide range of applications, including photography, film production, advertising, medical imaging, and content creation, where precise control over visual details is essential. In recent years, text-based image editing has emerged as a powerful and intuitive approach, enabling users to modify images through natural language instructions [Couairon et al. 2023; Zhang et al. 2023a]. While effective, relying solely on

text is often insufficient, as language may struggle to capture fine-grained visual nuances, spatial relationships, or stylistic intent [Li et al. 2023; Mou et al. 2024]. As a result, complementary techniques have been developed to bridge this gap, such as image composition methods that combine multiple visual elements [Song et al. 2023; Yang et al. 2023], sketch-based editing that provides explicit structural guidance [Mikaeili et al. 2023], and frameworks like ControlNet [Zhang et al. 2023b] that incorporate additional conditioning signals. Together, these approaches enhance controllability and enable more accurate and expressive image editing workflows.

An alternative and powerful editing approach is image analogy, which enables the transfer of transformations from one example to another in a structured way [Chen et al. 2025; Hertzmann et al. 2001; Šubrťová et al. 2023]. In this setting, a pair of images (A) and (A') represents a specific edit or transformation—such as a change in style, texture, or geometry. The goal is then to apply the same transformation to a new input image (B), producing a corresponding output (B') that reflects the edits observed between (A) and (A'). In other words, the system learns the mapping from (A) to (A') and generalizes it to (B), allowing consistent and controllable edits without requiring explicit textual or manual specification. This makes image analogy particularly useful for tasks where visual changes are easier to demonstrate than to describe. A text prompt or reference image specifies the category of an edit, while a before/after pair captures its exact visual form—including style, shape, size, placement, magnitude, and subject-specific interactions. These relative properties are difficult to convey through text or a single reference image. For example, in Fig. 1, text can specify adding an “SIA” logo but not its exact font; the demonstration pair directly provides this visual detail, enabling its transfer to the query subject.

However, prior image analogy methods typically assume that all changes observed between (A) and (A') should be fully transferred to the target image (B). This all-or-nothing behavior limits their flexibility in practical scenarios, where users may only want to apply a subset of the edits. For example, one might wish to transfer attributes such as a smile or the presence of sunglasses from (A') while preserving the original pose, lighting, or background of (B). By entangling multiple transformations into a single mapping, these approaches make it difficult to selectively control which aspects are propagated, thereby restricting their applicability in more nuanced and user-driven editing tasks.

To address this limitation, we introduce a new problem setting termed *Selective Image Analogy*, where the transformation from (A) to (A') may involve multiple edits, but only a subset of these edits is transferred to a target image (B) (see Fig. 1). Crucially, the selection of which edits to apply is guided by text, allowing users to specify their intent at a high level while still leveraging the rich visual cues embedded in the example pair. This formulation combines the strengths of text-based editing—flexibility and semantic control—with the expressiveness of image analogy, enabling more precise and user-driven manipulation of visual content.

To solve this problem, we design SIA so that the same demonstration pair (A, A') can support different user-selected outputs. The model first learns the standard analogy behavior, transferring the full transformation shown between A and A' to a new query image B . We then make this behavior selective through an edit-token

suppression mask, which marks the edits the user does not want to transfer. This mask controls suppression at two aligned points: during denoising, it applies a token-gated LoRA only to the suppressed edit tokens; before denoising, it weakens the corresponding text embeddings along edit directions estimated from positive and negative prompt pairs. This allows SIA to suppress unwanted edits that remain visible in A' , while preserving the edits selected by the user without requiring a new demonstration pair for each edit combination. We demonstrate the efficacy and generalizability of our method through extensive quantitative and qualitative evaluations, as well as a series of ablation studies that justify our design choices, all while maintaining a lightweight and training-efficient framework that requires no per-query optimization or new demonstration pairs. Therefore, our contributions are as follows:

- We introduce *Selective Image Analogy*, a new image analogy setting that, to our knowledge, is the first to allow a multi-edit before/after demonstration to be selectively transferred to a new image.
- We introduce a mask-aligned suppression mechanism that localizes suppression to user-selected edit phrases, enabling unwanted edits to be blocked even when they remain visible in the demonstration.
- We also curate multi-edit training examples with full-edit and partial-edit targets, allowing the model to learn both complete and selective transfer.

2 Related Works

2.1 Image editing.

Diffusion-based image editing was popularized by SDEdit [Meng et al. 2022], which edits an input image by adding noise and denoising it under a target condition. Later, instruction-based editors made this process more accessible by allowing users to describe edits with natural-language commands [Brack et al. 2024; Brooks et al. 2023; Kawar et al. 2023; Mokady et al. 2023].

Since text alone is often not enough for precise control, recent works incorporate additional signals such as structural guidance with ControlNet [Zhang et al. 2023b], reference-based conditioning [Ye et al. 2023], and attention-sharing mechanisms [Alimohammadi et al. 2025; Cao et al. 2023; Mikaeili et al. 2026; Xu et al. 2024]. More recently, transformer-based diffusion and flow models have shown strong versatility across local and global editing tasks [Labs et al. 2025; Wu et al. 2025]. Our work builds on this progress, but moves toward selective visual analogies, where the edit is given by a before/after demonstration and only the user-selected parts of that demonstration are transferred.

2.2 Image analogy.

In image analogy, the user demonstrates an edit with a before/after example, and the edit is transferred to another image, which shares similarity with the example pair. Classical Image Analogies [Hertzmann et al. 2001] introduced the general formulation of transferring the edit through example-based local visual matching. Deep

Image Analogy [Liao et al. 2017] extended this idea to semantically related image pairs by matching deep features, enabling visual attribute transfer across large appearance gaps, while Structural Analogy [Benaïm et al. 2021] learns a generative mapping between the structures of two images from a single pair. With the advent of stronger generative models such as diffusion models, image analogy was revisited using noise inversion techniques guided by the CLIP embedding space [Šubrtová et al. 2023]. Alternative approaches leveraged visual prompting and in-context learning, where example pairs are placed together with the query image as visual context [Bar et al. 2022; Gu et al. 2024; Sun et al. 2023; Wang et al. 2023]. The intuition is that, when these images are given jointly to a diffusion transformer, the model can compare them through attention and infer the transformation to apply, without optimizing a separate model for each test example.

Building on this direction, Edit Transfer [Chen et al. 2025] learns visual relations from source-target editing examples and applies them to new query images. LoRWeB [Manor et al. 2026] improves generalization by dynamically composing a basis of LoRA modules for different analogy types. VIRAL [Li et al. 2026] further studies visual in-context reasoning with diffusion transformers using role-aware multi-image conditioning and expert mixing. These methods show that visual demonstrations can capture transformations that are difficult to express with language alone, including spatial, non-rigid, and compositional edits. However, they generally transfer the demonstrated transformation as a whole. In contrast, our setting considers demonstrations where $A \rightarrow A'$ contains multiple edits, and the user may want to transfer only a selected subset while keeping the same demonstration fixed.

2.3 Controllable image editing.

Our work broadly falls within the domain of controllable image editing, where users are given more direct control over which visual changes are applied and how strongly they appear. Concept Sliders [Gandikota et al. 2024] learns LoRA sliders for visual concepts, allowing continuous control over semantics. TARA [Peng et al. 2026] uses token masks to reduce interference between LoRA modules in multi-concept personalization. SliderEdit [Zarei et al. 2025] similarly restricts LoRA updates to selected instruction tokens, enabling fine-grained control over different parts of a multi-instruction edit.

Another line of work controls generation and editing through meaningful directions in learned representation spaces. Early methods study such directions in GAN latent spaces [Shen et al. 2020] or use CLIP guidance to manipulate StyleGAN representations [Patashnik et al. 2021]. More recent diffusion and flow-based methods apply similar ideas to text embeddings and intermediate activations. Continuous Subject-Specific Attribute Control [Baumann et al. 2025] estimates CLIP directions from contrastive prompts. SHIFT [Konovalova et al. 2026] builds steering vectors from paired positive and neutral prompts and applies them to intermediate activations in DiT-based flow models, while Diffusion Sliders [Ekin and Gandelsman 2026] derives text-encoder directions from contrastive prompt pairs for controllable steering.

These works control visual changes through token-level mechanisms, adapters, and representation-space directions, and our proposed method shares this spirit. However, applying these techniques to selective image analogy is non-trivial: the edit to suppress is not absent from the input—it remains fully visible in A' —so the model cannot simply ignore it. Text removal alone is insufficient, since the visual demonstration actively reintroduces the suppressed edit signal. This helps suppression operate at two aligned stages: weakening the text-conditioning signal before denoising and gating the adapter updates during denoising, both localized to the specific edit tokens. Our work is therefore the first to make a single visual analogy demonstration selectable, allowing subset of edits to be transferred from the same fixed before/after pair.

3 Preliminaries

Flow Models. Flow-based generative models [Lipman et al. 2023; Liu et al. 2023; Tong et al. 2024] learn a velocity field that transports a noisy latent sample toward data distribution. Let x_0 denote latent of clean data and $x_1 \sim \mathcal{N}(0, I)$ denote a noise sample. A noised latent at time t is obtained by interpolation,

$$z_t = (1 - t)x_0 + tx_1, \quad (1)$$

and the model is trained to predict the velocity from x_0 to x_1 . In the *conditional* setting, the velocity network also receives a text prompt c and visual conditioning input y . The training objective is

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, x_0, x_1, y, c} [\|v_\theta(z_t, t, y, c) - (x_1 - x_0)\|_2^2]. \quad (2)$$

FLUX.2-Klein backbone. FLUX.2-Klein [Black Forest Labs 2025] follows a multimodal diffusion transformer design in which image and text tokens are processed jointly. Images are first encoded into latent tokens by a VAE, while the text prompt is encoded by a Qwen3 text encoder [Yang et al. 2025]. The transformer f_θ then denoises the target latent conditioned on both the visual stream and the text stream. This architecture is well-suited for our setting because selective image analogy requires the model to reason jointly over the visual demonstration (A, A'), the target image B , and the text tokens that name the edits.

Low-rank adaptation. LoRA [Hu et al. 2022] adapts a large pre-trained model efficiently without updating all of its weights. Given a frozen linear layer with weight matrix $W_0 \in \mathbb{R}^{m \times n}$, LoRA represents the trainable update as a low-rank decomposition $\Delta W = BA$, where $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and $r \ll \min(m, n)$. The adapted layer is then written as:

$$W = W_0 + \frac{\alpha}{r} BA, \quad (3)$$

where α controls the scale of the update. Since only A and B are trained, LoRA preserves the pre-trained model while allowing efficient task-specific adaptation.

4 Method

We propose SIA, a framework for *Selective Image Analogy* using a visual before/after demonstration. Given a source image A , its edited version A' , and a target image B , our goal is to transfer only the user-selected edits from the change $A \rightarrow A'$ onto B to produce B' , as illustrated in Figure 1. The transformation from A to A' may contain multiple edits $\mathcal{E} = \{e_1, e_2, \dots, e_N\}$, such as a pose change, an

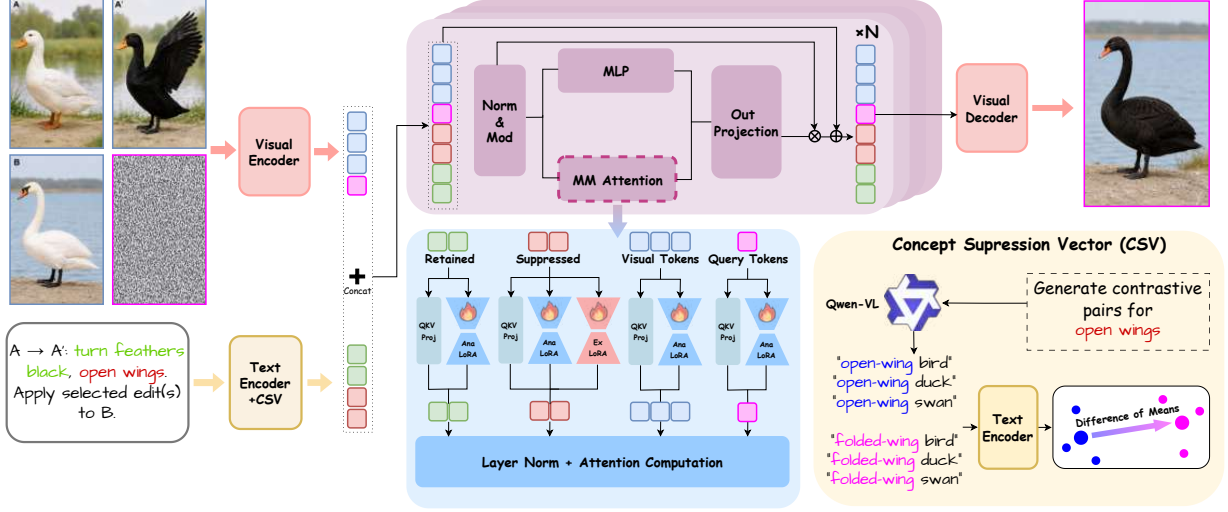


Fig. 2. **Overview of our selective image analogy pipeline.** Given a demonstration pair (A, A') containing multiple edits and a query image B , the model transfers only the user-selected edits to produce B' . In this example, **green textual tokens** indicate retained edits that should be applied to B , while **red textual tokens** indicate suppressed edits that should not be transferred, even though they are present in A' . The visual analogy guides the overall edit transfer, while the colored textual tokens specify which edits are retained or suppressed using selective token suppression and the Concept Suppression Vector (CSV).

expression change, a style transfer, or an object addition/removal. However, user may specify a subset $\mathcal{R} \subseteq \mathcal{E}$ to retain, leaving the suppressed edits $\mathcal{S} = \mathcal{E} \setminus \mathcal{R}$ to be blocked in the output $B'_{\mathcal{R}}$.

Unlike standard image analogy [Hertzmann et al. 2001], which transfers the full demonstrated transformation, SIA keeps the same before/after pair fixed while allowing selective control over which edits are applied. This setting differs from ordinary text-guided editing because the model also receives the visual pair (A, A') as conditioning: since A' already contains all demonstrated edits, simply removing an edit from the prompt is insufficient—the model can still infer and transfer it from A' . Rather than constructing a separate demonstration pair for each possible subset (which would be impractical), we use one full demonstration pair and control which edits are transferred at inference time via a token-level suppression mask that marks phrases corresponding to suppressed edits; SIA first learns to transfer all demonstrated edits and then applies this mask to selectively block them, as described below.

4.1 Full Transfer Model

Before introducing selective control, we first build the dataset and train the model to perform full visual analogy, both of which we describe in the following.

Dataset. Since no existing dataset supports selective multi-edit analogy, we construct one tailored to our task. Images are organized by *edit recipes*, each specifying a set of edits $\mathcal{E} = \{e_1, \dots, e_N\}$ with $N \in \{2, 3, 4\}$, where each group shows the same subject with different combinations of edits applied. For each subject in a recipe, we store the original image I^ϕ , the fully edited image $I^\mathcal{E}$, and partial images $I^\mathcal{R}$ for all retained subsets $\mathcal{R} \subseteq \mathcal{E}$. Training tuples are formed by pairing two subjects a and b from the same recipe, setting $A = I_a^\phi$, $A' = I_a^\mathcal{E}$, $B = I_b^\phi$, with targets $B'_\mathcal{E} = I_b^\mathcal{E}$ and $B'_\mathcal{R} = I_b^\mathcal{R}$ supervising full and selective transfer respectively.

The two-edit group covers pairwise combinations of object addition, style change, and background change, built by augmenting a subset of RelationAdapter [Gong et al. 2025] with a second edit synthesized via FLUX.2-Klein. The three- and four-edit groups are constructed by sequentially applying edits to source images, e.g. $I^\phi \rightarrow I^{\{e_1\}} \rightarrow I^{\{e_1, e_2\}} \rightarrow I^\mathcal{E}$, covering combinations of object additions, style changes, and background changes. All samples are filtered with a vision-language model to verify that desired edits are present and excluded edits are absent. In total, the dataset contains 1606 training samples drawn from 238 edit settings and 239 test cases, covering 2–4 simultaneous edits over both creature and non-creature subjects. Further details are provided in the supplementary material (Sec. E).

AnaLoRA. Given a demonstration pair (A, A') and a query image B , the model learns to transfer all demonstrated edits $\mathcal{E} = \{e_1, \dots, e_N\}$ and apply them to B , producing the full-edit output $B'_\mathcal{E}$. We utilize FLUX.2-Klein and apply a lightweight LoRA adapter, which we call *AnaLoRA*, trained to internalize the complete demonstrated transformation before any selective suppression is applied.

We adopt the in-context editing formulation of [Chen et al. 2025], based on a four-image tuple (A, A', B, B') representation. The latents of A , A' , and B are used as visual conditioning inputs, while B' denotes the target image to be generated. Let z_A , $z_{A'}$, z_B , and $z_{B'_\mathcal{E}}$ denote the latent tokens of the source image, edited source image, query image, and full-edit target, respectively. We pack the conditioning latents as $z_{\text{cond}} = [z_A; z_{A'}; z_B]$, where $[\cdot; \cdot]$ denotes concatenation along the token dimension. The visual conditioning is paired with a structured analogy prompt that names all demonstrated edits:

$\mathcal{P}(\mathcal{E}) = \text{"Image 1 is the original image."}$
 $\text{"Image 2 is the edited version of Image 1."}$
 $\text{"The edits are: } e_1, e_2, \dots, e_N."$
 $\text{"Apply the same edits to Image 3."}$

This prompt is encoded by the text encoder as $\phi_{\text{text}}(\mathcal{P}(\mathcal{E}))$ and injected into the MM-DiT text stream. The denoising transformer receives both the visual demonstration (A, A') and a text description of the edits to transfer. After this stage, the model has learned the default full-transfer behavior: given the complete demonstration, it transfers all edits in $\mathcal{E}$ to the query image. We then make this behavior controllable, so the same demonstration pair (A, A') can be used while selected edits are excluded.

4.2 Concept Suppression

Since suppressed edits remain visible in A' , simply removing them from the prompt is insufficient—the model can still infer and transfer them from the visual demonstration. Our goal is therefore to suppress only the excluded edits while keeping the demonstration pair fixed. We address this by conditioning selective transfer on a suppression mask. Given the user-selected retained set $\mathcal{R} \subseteq \mathcal{E}$, we define the suppressed set as $\mathcal{S} = \mathcal{E} \setminus \mathcal{R}$. Let $\tau(e_i) \subseteq \{1, \dots, T\}$ denote the set of token positions of the phrase describing edit e_i , where T is the text sequence length. We construct a binary token mask $M(\mathcal{S}) \in \{0, 1\}^T$ with $M_j(\mathcal{S}) = 1$ if $j \in U_{e_i \in \mathcal{S}} \tau(e_i)$ and 0 otherwise. Thus, $M(\mathcal{S})$ marks the edit phrases that should be excluded from transfer, while leaving the unmasked edit phrases available to guide generation. We use the same mask in the two suppression operations described next. ExLoRA changes how the denoising transformer handles the masked edit tokens during denoising, while CSV weakens the same edit phrases in the prompt embeddings before denoising.

ExLoRA. To make the suppression mask affect denoising, we introduce *ExLoRA*, a token-gated LoRA [Zarei et al. 2025] that applies its low-rank update only to the text tokens marked by the suppression mask—those corresponding to demonstrated edit phrases that should be excluded from the output. While AnaLoRA (full edit) remains active for the full-transfer analogy behavior, ExLoRA handles the edit phrases marked by the suppression mask. It does not learn a separate analogy model. Instead, it changes how the denoising transformer handles the masked edit phrases, allowing the model to use the same full demonstration while suppressing only the excluded edits. Formally, let h_j^ℓ denote the hidden representation of text token j at layer ℓ , and let W^ℓ be a frozen linear projection in the text stream. While a standard LoRA update applies the same residual to every token, ExLoRA gates this residual by the suppression mask:

$$\tilde{h}_j^\ell = \left(W^\ell + M_j(\mathcal{S}) \frac{\alpha}{r} B_s^\ell A_s^\ell \right) h_j^\ell, \quad j = 1, \dots, T. \quad (5)$$

Here, A_s^ℓ and B_s^ℓ are the selective LoRA weights, r is the rank, and α is the LoRA scaling factor. When $M_j(\mathcal{S}) = 0$, token j is processed by the ordinary text stream. When $M_j(\mathcal{S}) = 1$, the selective

LoRA residual is applied, so the model treats the corresponding edit phrase as suppressed.

Suppression is therefore localized to the masked edit phrase. The model does not require a new demonstration pair, a separate prompt for each subset, or a query-specific adapter. Instead, it receives the same full analogy input together with a token mask specifying which edit descriptions should be suppressed.

Concept Suppression Vector While ExLoRA modifies how the denoising transformer responds to masked tokens, the prompt embeddings still encode all edits, including suppressed ones. To remove this residual signal in the text conditioning, we apply a token-local Concept Suppression Vector (CSV) before denoising.

For each edit e_i , we estimate an edit direction d_i in the text-encoder embedding space. We use contrastive prompt pairs in which the edit is present or absent. As a result, given positive and negative prompt pairs $\{(p_n^+, p_n^-)\}_{n=1}^N$, we encode them with the same text encoder used by FLUX.2-Klein and pool the hidden states corresponding to the target edit phrase. Let u_n^+ and u_n^- denote the pooled positive and negative embeddings. We compute

$$d_i = \frac{\bar{u}_i^+ - \bar{u}_i^-}{\|\bar{u}_i^+ - \bar{u}_i^-\|_2}, \quad \bar{u}_i^+ = \frac{1}{N} \sum_{n=1}^N u_n^+, \quad \bar{u}_i^- = \frac{1}{N} \sum_{n=1}^N u_n^-. \quad (6)$$

The resulting direction is computed once for each edit phrase and cached. At inference time, if edit e_i belongs to the suppressed set $\mathcal{S}$, we apply CSV only to the token positions corresponding to that edit phrase. For each token embedding h_j with $j \in \tau(e_i)$, we remove its component along the edit direction:

$$h_j \leftarrow h_j - \lambda_i (h_j^\top d_i) d_i, \quad j \in \tau(e_i), \quad (7)$$

where λ_i controls the suppression strength. Tokens corresponding to retained edits and general analogy instructions are left unchanged.

CSV and ExLoRA operate at different stages using the same mask: CSV weakens suppressed edit tokens in the prompt embeddings before denoising, while ExLoRA modifies how the transformer processes them during denoising, together reducing leakage of suppressed edits into the output.

4.3 Training and Inference

We now describe the training procedure and inference process. Training uses both full-transfer and selective-suppression examples. The learned adapters, AnaLoRA and ExLoRA, are optimized with the flow-matching loss, while CSV is computed separately from contrastive text pairs and applied only at inference time.

We use the same visual conditioning latents throughout training,

$$z_{\text{cond}} = [z_A; z_{A'}; z_B],$$

where z_A , $z_{A'}$, and z_B are the latent tokens of the source image, edited source image, and query image, respectively.

In both training cases, the model receives the same visual conditioning input (A, A', B) and the same full analogy prompt $\mathcal{P}(\mathcal{E})$. The difference is whether the suppression mask is inactive or active, and whether the target contains all demonstrated edits or only a retained subset.

Full-transfer step. In a full-transfer step, the model receives the demonstration pair (A, A'), the query image B , and the full prompt

Table 1. Quantitative comparison with visual analogy baselines. VLM scores are reported using Qwen3-VL-32B-Instruct and Gemma 3. Ret. Acc. measures retained-edit quality, Pres. measures preservation of B , and Recall measures retained-edit recall. CLIP-Dir Image is the image-image directional metric, while CLIP-Dir Retained measures alignment with the retained edit direction.

Method	Qwen3-VL-32B			Gemma 3			CLIP	
	Ret. Acc. $\uparrow$	Pres. $\uparrow$	Recall $\uparrow$	Ret. Acc. $\uparrow$	Pres. $\uparrow$	Recall $\uparrow$	Dir. Image $\uparrow$	Dir. Ret. $\uparrow$
RelationAdapter [Gong et al. 2025]	6.74	9.93	0.782	8.09	9.48	0.941	0.386	0.144
VisualCloze [Li et al. 2025]	7.24	8.62	0.856	8.14	9.28	0.947	0.464	0.152
Edit Transfer [Chen et al. 2025]	2.31	9.96	0.176	5.75	9.58	0.623	0.102	0.023
LoRWeB [Manor et al. 2026]	6.59	9.83	0.736	8.12	9.51	0.926	0.402	0.155
Ours	8.01	9.75	0.892	8.65	9.57	0.952	0.543	0.171

$\mathcal{P}(\mathcal{E})$. No edit is suppressed, so the suppression set is empty and the token mask is zero:

$$\mathcal{S} = \emptyset, \quad M(\mathcal{S}) = \mathbf{0}. \quad (8)$$

The training target is the full-edit image $B'_{\mathcal{E}}$. Given Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$ and flow time $t \in [0, 1]$, we form

$$z_t = (1 - \sigma_t)z_{B'_{\mathcal{E}}} + \sigma_t\epsilon, \quad v_{\mathcal{E}}^* = \epsilon - z_{B'_{\mathcal{E}}}. \quad (9)$$

The full-transfer prediction is

$$\hat{v}_{\mathcal{E}} = f_{\theta, \text{AnaLoRA}}(z_t, z_{\text{cond}}, \phi_{\text{text}}(\mathcal{P}(\mathcal{E})), t), \quad (10)$$

and we optimize

$$\mathcal{L}_{\text{full}} = \mathbb{E}_{t, \epsilon} \left[\|\hat{v}_{\mathcal{E}} - v_{\mathcal{E}}^*\|_2^2 \right]. \quad (11)$$

This step teaches AnaLoRA the default analogy behavior: given the full demonstration, the model should transfer all demonstrated edits to the query image.

Selective-suppression step. In a selective-suppression step, the model still receives the full demonstration (A, A') , the query image B , and the full prompt $\mathcal{P}(\mathcal{E})$ naming all edits. We sample a retained subset $\mathcal{R}$, construct the suppression mask $M(\mathcal{S})$ over the excluded edit phrases, and train against the partial target $B'_{\mathcal{R}}$, which contains only the retained edits.

For this target, we form

$$z_t = (1 - \sigma_t)z_{B'_{\mathcal{R}}} + \sigma_t\epsilon, \quad v_{\mathcal{R}}^* = \epsilon - z_{B'_{\mathcal{R}}}. \quad (12)$$

The selective prediction is

$$\hat{v}_{\mathcal{R}} = f_{\theta, \text{AnaLoRA}, \text{ExLoRA}}(z_t, z_{\text{cond}}, \phi_{\text{text}}(\mathcal{P}(\mathcal{E})), M(\mathcal{S}), t). \quad (13)$$

We optimize the selective flow-matching loss

$$\mathcal{L}_{\text{sel}} = \mathbb{E}_{t, \epsilon} \left[\|\hat{v}_{\mathcal{R}} - v_{\mathcal{R}}^*\|_2^2 \right]. \quad (14)$$

This supervision is counterfactual: the input shows and names the full transformation, but the target removes only the masked edits. Thus, the model learns to control which edits are transferred by changing the mask, without needing an input for edit subsets.

Inference. At inference time, the user provides one demonstration pair (A, A') , one query image B , and the edits to retain, $\mathcal{R}$. We define the suppressed edits as $\mathcal{S} = \mathcal{E} \setminus \mathcal{R}$. We keep the full visual conditioning (A, A', B) and the full prompt $\mathcal{P}(\mathcal{E})$ unchanged. We then construct the token mask $M(\mathcal{S})$ over the suppressed edits.

The same mask controls both parts of the selective-conditioning mechanism. It first determines where CSV weakens the prompt

embeddings using Eq. 7. The same token positions then activate ExLoRA inside the text stream through Eq. 5. The denoising process then generates $B'_{\mathcal{R}}$, which should preserve the retained demonstrated edits while excluding the suppressed ones. Thus, inference uses one fixed demonstration pair and changes only the suppression mask over the edit tokens and no subset-specific examples or query-specific optimization are required.

5 Experiments

We evaluate SIA on multiple demonstrated edits using our Multi-Edit Analogy Dataset. Our goal is to evaluate selective transfer quality, the contribution of ExLoRA and CSV over prompt-based suppression, and robustness across edit types and multi-edit settings. Two user studies evaluating the same criterion are reported in Sec. B.

5.1 Experimental Setup.

We evaluate SIA on FLUX.2-Klein [Black Forest Labs 2025] in concat-mode, with A , A' , and B as conditions. Both AnaLoRA and ExLoRA use rank $r=16$, learning rate 5×10^{-5} , and up to 50K steps, with batch sizes of 16 and 24 respectively. At inference, we use 30 denoising steps with ExLoRA scale 1.0. Each CSV direction is estimated from 64 prompt pairs per edit, cached once, and applied with suppression strength selected from $\{0, 75, 150, 225, 300\}$. We compare with VisualCloze [Li et al. 2025], RelationAdapter [Gong et al. 2025], LoRWeB [Manor et al. 2026], and Edit Transfer [Chen et al. 2025].

5.2 Qualitative Results.

Fig. 5 shows results of SIA across a set of selective image analogy tasks. The examples cover local object additions, facial expression changes, background replacements, and style transfer. Existing visual analogy methods often transfer the full transformation from A' , including edits selected for suppression, or preserve B by making only weak changes. In contrast, SIA better separates the demonstrated edits: it preserves the retained transformations, suppresses the user-selected edit, and maintains the identity and overall structure of the query image B . These results show that our method can treat a multi-edit demonstration as a controllable set of edit factors rather than a single entangled transformation.

Comparison to Baselines. Since existing visual analogy methods are designed for full edit transfer rather than selective transfer, we adapt them to our setting by providing the same input triplet (A, A', B) . For the input text prompt, we provide the retained-edit

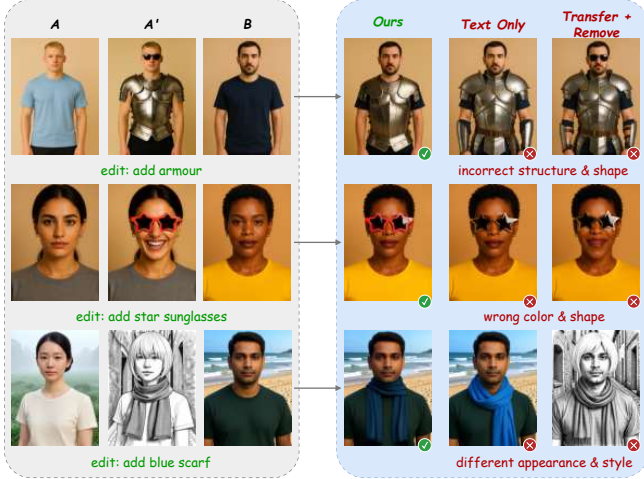


Fig. 3. **Advantage of demonstration-based specification.** Text conveys the category of an edit but not its exact form. In each row, the demonstration specifies the exact visual form of the edit, including its structure, shape, and colour, which text-based editing fails to reproduce, while SIA matches the demonstrated transformation.

prompt. This gives each baseline access to the user-desired edit subset, while keeping the visual demonstration fixed. The comparison, therefore, tests whether each method can avoid copying suppressed edits from A' , where the full transformation remains visually present. Fig. 4 shows qualitative comparisons with baselines.

5.3 Quantitative Evaluation.

We report both CLIP-based and VLM-based metrics. For CLIP-based evaluation, we report two directional metrics. *CLIP-Dir Image* is the image-image directional metric that measures agreement with the demonstrated image-space transformation. *CLIP-Dir Retained* measures whether the generated output moves in the direction of the retained edit descriptions, making it suitable for selective transfer.

For semantic evaluation, we use a VLM-based protocol, following the recent use of vision-language models for assessing image editing quality and human-aligned preferences [Huang et al. 2025]. Qwen3-VL-32B-Instruct [Bai et al. 2025] and Gemma 3 [Gemma Team et al. 2025] are used as independent judges. Each VLM is given the demonstration pair (A, A') , the query image B , the generated output $\hat{B}'$, and the lists of retained and suppressed edits. We report three VLM metrics. *Retained Accuracy* measures the quality of applying the retained edits. *Preservation* measures whether the output maintains the identity, layout, and unrelated content of B . *Edit Recall* measures the fraction of retained edits detected in the output.

Tab. 1 compares SIA against recent visual analogy baselines. SIA achieves the highest retained-edit accuracy and edit recall under both VLM judges, and also obtains the best CLIP directional scores. We note that methods such as Edit Transfer can obtain high preservation because they often make only weak changes or leave the query image close to unchanged. However, preservation alone is not sufficient for selective image analogy: a successful method must

Table 2. Quantitative ablation of the suppression components on the test set. Supp. measures the absence of suppressed edits in the output. AnaLoRA is active in all variants.

ExLoRA	CSV	Ret. Acc. $\uparrow$	Supp. $\uparrow$	Pres. $\uparrow$	Avg. $\uparrow$
No	No	9.07	1.93	9.00	6.67
No	Yes	8.81	4.93	9.16	7.63
Yes	No	7.95	7.81	9.77	8.51
Yes	Yes	8.01	7.85	9.75	8.54

also apply the retained edits accurately, which is reflected by Retained Accuracy and Edit Recall.

Comparison to text-based only editing. Given the strength of text- and reference-conditioned editors, it is worth examining if a demonstration pair is necessary. Fig. 3 demonstrates that SIA better captures the desired edit while avoiding unwanted changes. Using FLUX.2-Klein-9B with a retained-edit text instruction, the *text-only* baseline struggles to suppress unwanted edits, while *Full Transfer + Removal* first copies all demonstrated edits and then attempts to remove the suppressed ones. Additional quantitative and qualitative comparisons with text-only baselines and commercial tools such as Nano Banana are provided in the supplementary material.

5.4 Ablation Study.

We ablate the two selective suppression components of SIA in Tab. 2 and Sec. F. In addition to Retained Accuracy and Preservation, we report a VLM-based Suppression score (*Supp.*), measuring whether suppressed edits are absent from the output. AnaLoRA remains active in all variants as the base visual-analogy capability. Without ExLoRA and CSV, the model transfers nearly the full transformation from A' . CSV alone weakens the suppressed edit’s text signal but lacks a learned suppression pathway, while ExLoRA alone provides token-local suppression but leaves residual semantic evidence in the prompt embeddings. Combining both yields the strongest suppression and overall performance. The slight drop in Retained Accuracy reflects the trade-off between suppression strength and retained-edit fidelity, while visual ablations confirm their complementary roles.

6 Limitations, Future Work and Conclusion

We introduced Selective Image Analogy, a novel problem setting that, to our knowledge, is the first to enable selective transfer from a multi-edit visual demonstration. Unlike prior image analogy methods that treat the demonstrated transformation as an indivisible whole, SIA allows a user to choose which edits from a single before/after pair are applied to a new query image, while suppressing the rest—even when the suppressed edits remain visible in the demonstration. To support this setting, we constructed a multi-edit analogy dataset and proposed a mask-aligned suppression mechanism combining AnaLoRA for full-transfer learning, ExLoRA for token-gated suppression during denoising, and CSV for weakening residual edit signals in the text embeddings before denoising. Extensive quantitative evaluations, qualitative comparisons, and a

user study confirm that SIA consistently outperforms existing visual analogy baselines on the selective transfer objective, demonstrating that a single demonstration pair can serve as a flexible, user-controllable source of edits rather than a fixed, all-or-nothing transformation.

While SIA currently supports up to a handful of demonstrated edits, it may struggle when A and A' differ substantially. The dataset also covers a limited range of edit types and scenes, leaving out fine-grained operations such as artifact removal and complex layouts. Scaling to a larger number of edits and broader edit diversity remains an interesting direction for future work. Another promising direction is compositional demonstration, where edits are drawn from multiple different before/after pairs and combined into a single selective transfer, allowing users to mix demonstrated edits across sources with controllable intensity rather than being constrained to a single (A, A') .

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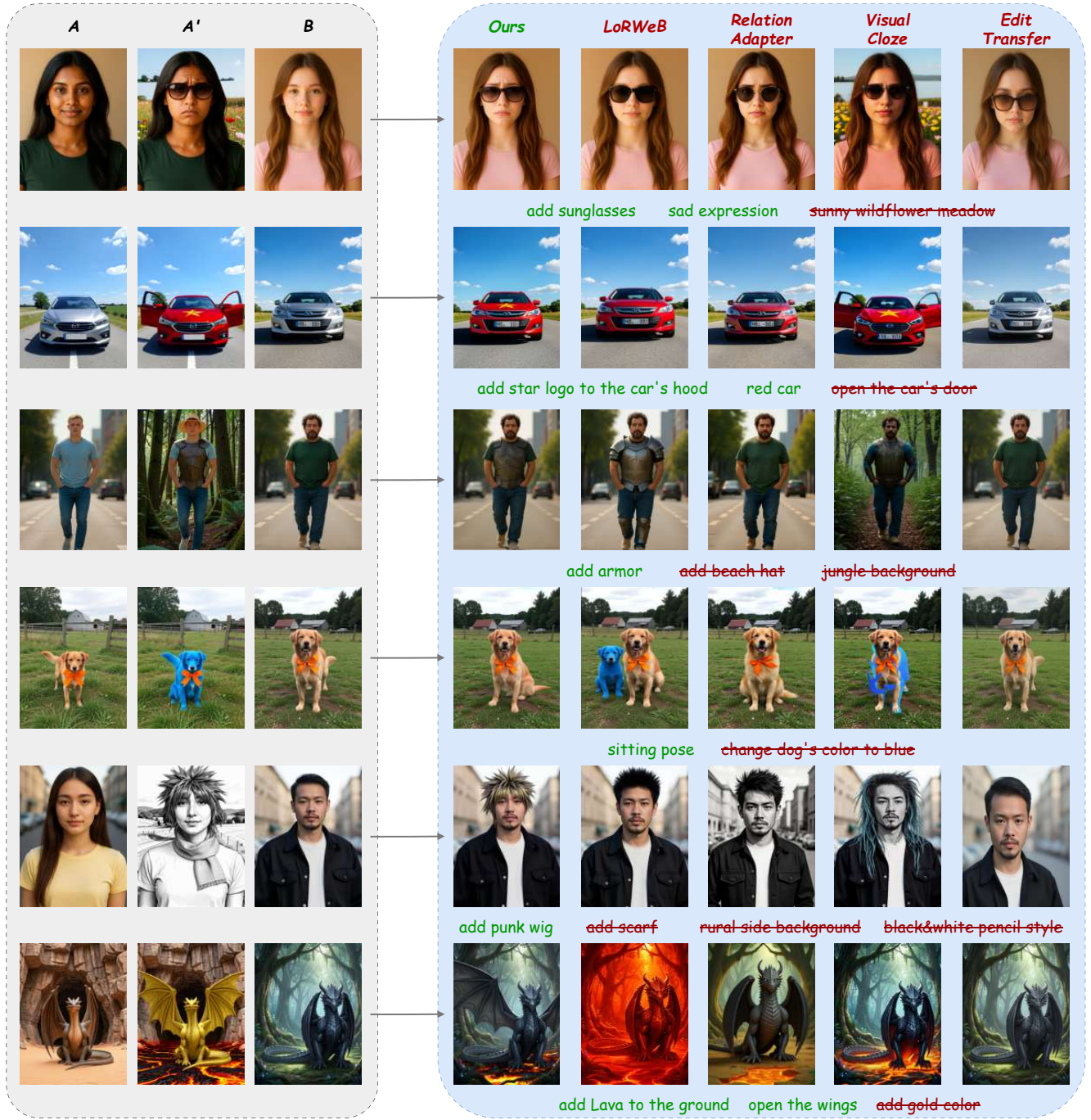


Fig. 4. **Visual comparison.** Qualitative comparisons with existing approaches demonstrate that only our method can selectively suppress undesired edits while preserving the input image and applying the intended modifications. In contrast, other methods either apply all edits simultaneously (e.g., Relation Adapter) or substantially alter the input image, such as the dog introduced by LoRWeB.

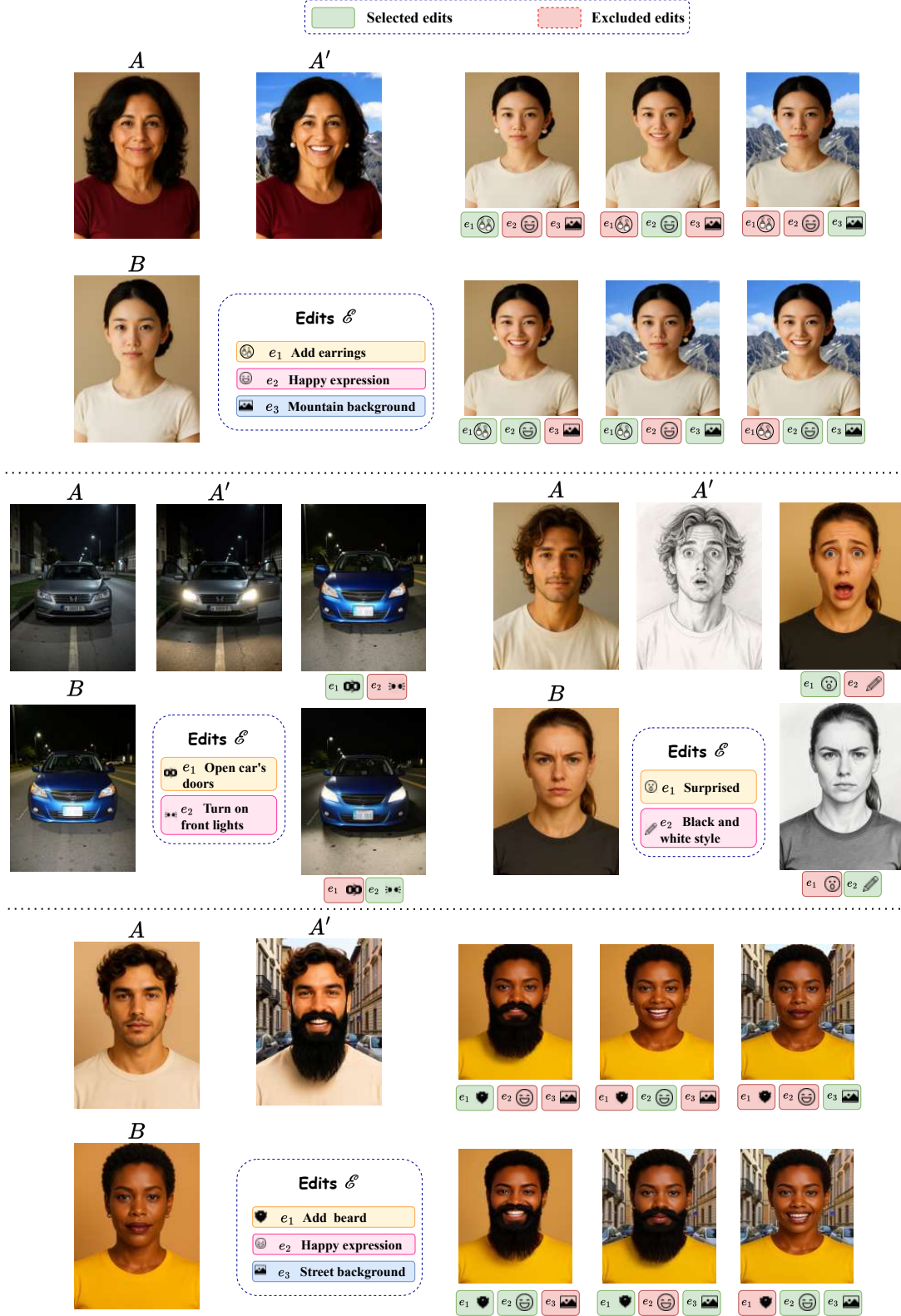


Fig. 5. **Qualitative results.** We provide multiple examples with different numbers of edits that can be switched to active or inactive. Our method is able to successfully control which edit needs to be applied.

A Implementation Details

Backbone. All experiments use 9B-parameter FLUX.2-Klein [Black Forest Labs 2025] as the diffusion backbone. The text stream is encoded by the Qwen3 text encoder packaged with FLUX.2-Klein. The diffusion pipeline concatenates the hidden states from three Qwen3 layers ($\{9, 18, 27\}$) along the channel dimension to form the per-token prompt embedding consumed by the MM-DiT blocks. As described in the main paper, ExLoRA is injected into the text-stream projections of the MM-DiT blocks.

Adapters. AnaLoRA and ExLoRA are independent LoRA adapters with rank $r=16$ and dropout 0. We first train AnaLoRA with batch size 16, learning rate 5×10^{-5} , and up to 50K optimization steps on a mixture of full-transfer tuples from the two-, three-, and four-edit groups. We then merge the resulting checkpoint into the frozen backbone and train ExLoRA on top, using batch size 24 and the same learning rate. To keep ExLoRA close to the full-transfer behavior at the beginning of training, roughly one third of the optimization steps instead update AnaLoRA jointly with an anchoring weight of 0.1, with the suppression mask disabled. Both adapters are optimized with AdamW and gradient clipping at $\|g\|_2 \leq 1$. We use the flow-matching loss described in the main paper, with timesteps sampled uniformly from $[0, 1000]$.

Inference. At inference time, we run 28 denoising steps in concat mode. The three conditioning latents $z_A, z_{A'}, z_B$ are concatenated along the token dimension, and the model is not given a 2×2 grid. The Klein backbone does not support classifier-free guidance, so we do not pass a negative prompt. The ExLoRA scale is fixed at 1.0 for all reported numbers. Setting it to 0 gives the AnaLoRA-only baseline.

CSV pipeline. For every edit phrase, the steering direction is built once and cached. The pipeline follows the Diffusion-Sliders [Ekin and Gandelsman 2026] recipe, but uses the diffusion pipeline’s own text encoder so that the direction is defined in exactly the embedding space consumed by the denoiser:

- (1) For each edit, we prompt the same Qwen3 encoder in causal-LM mode to generate $N=64$ contrastive pairs $\{(p_n^+, p_n^-)\}$, where the target edit is the only varying attribute.
- (2) We re-encode every sentence with the same encoder to obtain the per-layer hidden states.
- (3) We mean-pool the hidden states at the token positions that name the target edit.
- (4) We store the unit-normalized difference of means $d_i = \frac{\bar{u}_i^+ - \bar{u}_i^-}{\|\bar{u}_i^+ - \bar{u}_i^-\|_2}$ on disk together with the generated pairs. The cache is keyed by sha1(edit phrase), so the same vector is reused across all queries that mention the same edit.

Cross-concept pair construction. A naive contrastive-pair construction uses only the target edit, for example “A person with a beard” versus “A person without a beard”. In practice, this produces directions that share a generic “edit-present vs. edit-absent” axis across different edits, with pairwise cosines in the 0.85–0.94 range. Suppressing one edit can then partially suppress another, and the per-edit mask in ExLoRA loses its intended meaning. To avoid this, we construct *cross-concept* pairs where every other edit in the same

Table 3. **User study results.** Participants rank the top three outputs according to selective analogy quality. Our method receives the highest average ranking score.

Method	Ave. Ranking Score $\uparrow$
RelationAdapter [Gong et al. 2025]	1.22
VisualCloze [Li et al. 2025]	0.81
Edit Transfer [Chen et al. 2025]	0.12
LoRWeB [Manor et al. 2026]	1.38
Ours	2.47

Table 4. **Text-baseline user study.** Participants rate each output for Fidelity on a 0–10 scale and choose the single best overall result per sample. SIA achieves the highest Fidelity and is most often chosen as best overall.

Method	Fidelity $\uparrow$	Overall (%) $\uparrow$
Text Only	7.00	30.1
Full Transfer + Removal	5.78	4.4
SIA (Ours)	8.86	65.4

recipe is treated as a *companion* and held fixed across the positive and negative sentences. For example, for add beard with companion add sunglasses, we use “A person with a beard, wearing sunglasses” versus “A clean-shaven person wearing sunglasses”. Since both sides include the companion phrase, its contribution cancels in the difference of means, and the resulting direction better isolates the beard axis. To make the direction robust to whether the companion is present at inference time, we stratify the pairs across three regimes, each contributing roughly one third of the set: (A) the companion is present in both sentences, (B) the companion is absent in both sentences, and (C) the companion is not mentioned in either sentence. After this construction, the measured pairwise cosines between different edit directions are approximately 0.30, which is the regime where per-edit suppression behaves as intended.

Hardware and training time. All training runs were performed on a single RTX PRO 6000 WS GPU. Inference for one (A, A', B) tuple at 28 steps and 512×512 resolution takes under one minute. We will release the training configuration files and evaluation scripts to support reproducibility.

B User Study

Standard image-level metrics, such as LPIPS and CLIP similarity, do not fully capture the main requirement of selective image analogy, where the output must apply the retained edits, avoid the suppressed edit, and still preserve the query image B . We therefore conduct a user study focused on this task-specific criterion. In each trial, participants are shown the demonstration pair (A, A') , the query image B , the retained edits, the edit selected for suppression, and outputs from SIA and the baseline methods in random order. Participants are asked to rank the top three outputs according to how well they satisfy the selective analogy objective, considering

three factors jointly: applying the retained edits, avoiding the suppressed edit, and preserving the identity and structure of B . We collect responses from 37 participants over 15 examples and compare SIA against the baselines. For scoring, the first-ranked result receives 3 points, the second receives 2 points, the third receives 1 point, and unranked results receive 0 points. The results in Tab. 3 show that our method receives the highest average ranking score, confirming that SIA better matches the intended selective edit behaviour.

We additionally conduct a second user study that directly compares SIA against the *Text Only* and *Full Transfer + Removal* baselines, the strongest text-conditioned alternatives to demonstration-based transfer. In each trial, participants see the demonstration pair (A, A') , the query image B , the retained-edit and suppressed-edit lists, and the outputs of the three methods in random order. Participants rate the *Fidelity* on a 0–10 scale. *Fidelity* measures whether the retained edits match the demonstrated form in A' with the same style, shape, and colour, and not merely the same category of edit, where a low score means the retained edits are missing or bear no resemblance to A' , a mid-range score means the correct type of edit but a different instance (for example, sunglasses of a different style), and the maximum means every retained edit is present and matches the exact appearance in A' . After rating all three outputs, participants additionally select the single best overall result for that sample. We collect responses from 20 participants over 8 examples. As shown in Tab. 4, SIA attains the highest Fidelity score and is chosen as the best overall result in 65.4% of trials, compared with 30.1% for *Text Only* and 4.4% for *Full Transfer + Removal*. While text-based editing often produces the correct semantic edit, it fails to reproduce the demonstrated visual form, which is precisely what the Fidelity metric penalises.

C Comparison to text-based editing and commercial models.

A natural question is whether we even need the demonstration pair at all, given how strong text- and reference-conditioned editors have become. To check this, we compare SIA against three additional baselines on the same test set. (i) *Text Only* simply runs FLUX.2-Klein-9B with the retained-edit instruction as text, with no demonstration pair involved. (ii) *Full Transfer + Removal* first copies over every demonstrated edit, then tries to undo the suppressed ones with a text instruction. (iii) *Nano Banana* is a commercial black-box editor that takes the full triplet (A, A', B) along with the retained-edit instruction; qualitative comparisons against it are shown in Sec. D.

Results are reported in Tab. 5. All three baselines score high on Recall, meaning they generally produce the right *kind* of edit. But Recall only checks whether an edit happened. It does not check whether the edit looks like the one demonstrated in A' , matching its style, shape, size, placement, and scale. That is what Retained Accuracy measures, and this is where SIA pulls ahead of every baseline (Fig. 3). *Full Transfer + Removal* does even worse. Its removal step often changes parts of B that should have stayed the same, which lowers its Preservation score. These metrics can still miss

Table 5. Comparison with text-based editing baselines and a commercial multi-reference editor. Text-based baselines achieve high Recall but lower Retained Accuracy.

Method	Ret. Acc. $\uparrow$	Recall $\uparrow$	Pres. $\uparrow$
Text Only	7.43	0.917	9.68
Full Transfer + Removal	7.68	0.921	8.99
Nano Banana	7.36	0.885	9.55
SIA (Ours)	8.01	0.892	9.75

Table 6. Side-by-side VLM preference. The judge sees (A, A') , B , the retained/suppressed edit lists, and two outputs in random order.

Baseline	Ours Win (%) $\uparrow$	Baseline Win (%) $\downarrow$	Tie %
Text Only	56.9	17.6	25.5
Full Transfer + Removal	48.5	30.1	21.4

subtler visual mismatches, so we run a second check: a head-to-head comparison judged by Qwen3-VL-32B-Instruct (Tab. 6). For each pair, the judge sees (A, A', B) , the retained-edit description, and one output from SIA next to one from either *Text Only* or *Full Transfer + Removal*. The judge then picks the image that better applies the retained edit while suppressing the rest. Across both comparisons, SIA is chosen as better or tied roughly 70% of the time.

D SIA vs Nano Banana qualitative comparison

Beyond the analogy-based baselines evaluated in the main paper, a natural question is how SIA compares against strong commercial editors that accept multiple reference images directly. We therefore compare against Nano Banana, a black-box multi-reference editor, giving it the same demonstration pair (A, A') , the query image B , and the retained-edit instruction. Fig. 6 shows the results across a range of edits, including adding sunglasses, a crown, a logo, a beard, and lipstick. In every case, Nano Banana recognises the correct kind of edit but does not reproduce the specific form demonstrated in A' : the sunglasses and crown take on the wrong shape, the star logo is placed incorrectly, and the lipstick appears in the wrong colour. SIA, by contrast, matches both the type and the appearance of each demonstrated edit while leaving the identity and structure of B intact. This highlights a key advantage of demonstration-based transfer: text or reference conditioning alone conveys *what* to edit, but not the precise visual form the edit should take.

E Multi-Edit Analogy Dataset

In multi-edit image analogy, the model must transfer a demonstrated transformation from one subject to another, while also separating that transformation into its individual edit components. This requires supervision beyond standard image-editing datasets. For each subject, we need the original image, the fully edited image, and partial-edit images that correspond to different *subsets* of the demonstrated edits. Since no existing dataset provides this structure, we build a new **multi-edit analogy dataset** for our task. Fig. 7 shows the generation pipeline, and Fig. 8 shows the storage layout for each sample.

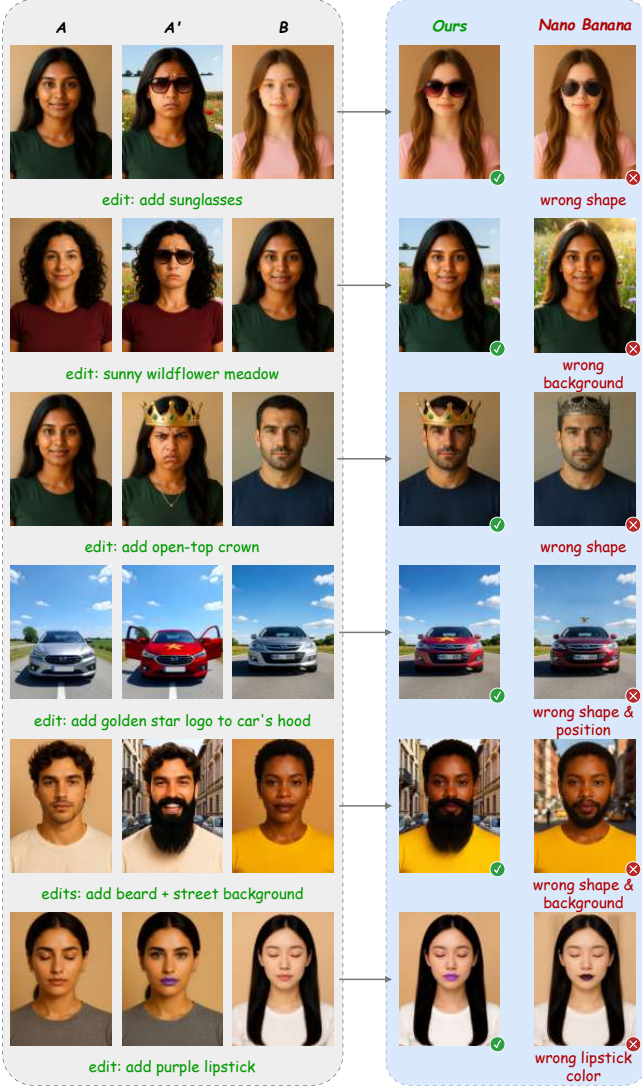


Fig. 6. **Visual comparison with Nano Banana.** Qualitative comparisons against Nano Banana, a commercial multi-reference editor, show that only SIA reproduces the demonstrated edit’s exact visual form. Nano Banana typically identifies the correct *category* of edit, but fails to match the specific appearance shown in A' : it renders sunglasses and the crown with the wrong shape, places the star logo in the wrong position, and produces an incorrect lipstick color. In contrast, SIA consistently matches both the type and the precise form of each demonstrated edit while preserving the identity and structure of B .

E.1 Edit recipes and storage format

Our dataset is organized around *edit recipes*. An edit recipe defines the set of edits applied to all subjects in a group, such as adding an object, changing the background, or changing the image style. We represent each recipe as a set of edits $\mathcal{E} = \{e_1, \dots, e_N\}$, where $N \in \{2, 3, 4\}$. For each subject in a recipe, we store the original image I^ϕ , the full-edit image $I^\mathcal{E}$, and the intermediate partial images $I^\mathcal{R}$

for the retained edit subsets $\mathcal{R} \subseteq \mathcal{E}$ that are reachable through the sequential synthesis process. As in the main paper, $\mathcal{R}$ denotes the retained edits and $\mathcal{S} = \mathcal{E} \setminus \mathcal{R}$ denotes the suppressed edits. Each edit also has a short natural-language description, which is used both for prompting and for constructing the contrastive pairs used by CSV (Sec. A).

Training tuples are formed by pairing two subjects a and b from the same recipe:

$$A = I_a^\phi, \quad A' = I_a^\mathcal{E}, \quad B = I_b^\phi.$$

For the query subject b , the full target $B'_\mathcal{E} = I_b^\mathcal{E}$ contains all demonstrated edits, while the selective target $B'_\mathcal{R} = I_b^\mathcal{R}$ contains only the retained edits. Together, these targets provide the supervision needed for selective analogy. The full target $B'_\mathcal{E}$ teaches the model to transfer the complete demonstrated transformation, while the selective target $B'_\mathcal{R}$ teaches the model how the same demonstration should be interpreted when the mask excludes the edits in $\mathcal{S}$.

Concretely, every leaf directory contains the source image I^ϕ , the reference images that visually anchor each individual edit (for example, the style reference, object reference, or target expression), the sequence of composite images corresponding to partial-edit subsets, and a prompt .json file that maps each composite to a short natural-language instruction. Fig. 8 summarizes this layout.

E.2 Two-edit data from RelationAdapter and style sources

The first part of our dataset contains two-edit analogies built from a subset of RelationAdapter [Gong et al. 2025]. RelationAdapter provides diverse single-edit transformations, such as changing character attributes, modifying pose, adding accessories, or altering local appearance. We treat each selected RelationAdapter pair as the first edit, and introduce a second edit by applying a style transformation defined by a small set of predefined style reference images. We use FLUX.2-Klein in style-transfer mode to synthesize both the style-only composite and the final two-edit composite. Each example therefore contains four images of the same subject: the original image, the RelationAdapter-edited image, the style-only image, and the final image that combines both edits. Since RelationAdapter provides most of the visual diversity, this subset is the main source of edit variation in our dataset.

E.3 Synthetic three-edit and four-edit data

The second part of our dataset contains more complex edit recipes with three or four edits. These examples are designed to expose the model to cases where multiple edits must be suppressed at the same time while the remaining edits are preserved. For these settings, we first generate source images with FLUX.2-Klein and then use the same model to synthesize the edited variants, including object additions, style changes, and background changes.

Starting from the original image I^ϕ , edits are applied sequentially. We first generate the single-edit image, then the two-edit image, and finally the full multi-edit image. For a three-edit recipe, this gives the progression

$$I^\phi \rightarrow I^{\{e_1\}} \rightarrow I^{\{e_1, e_2\}} \rightarrow I^\mathcal{E},$$

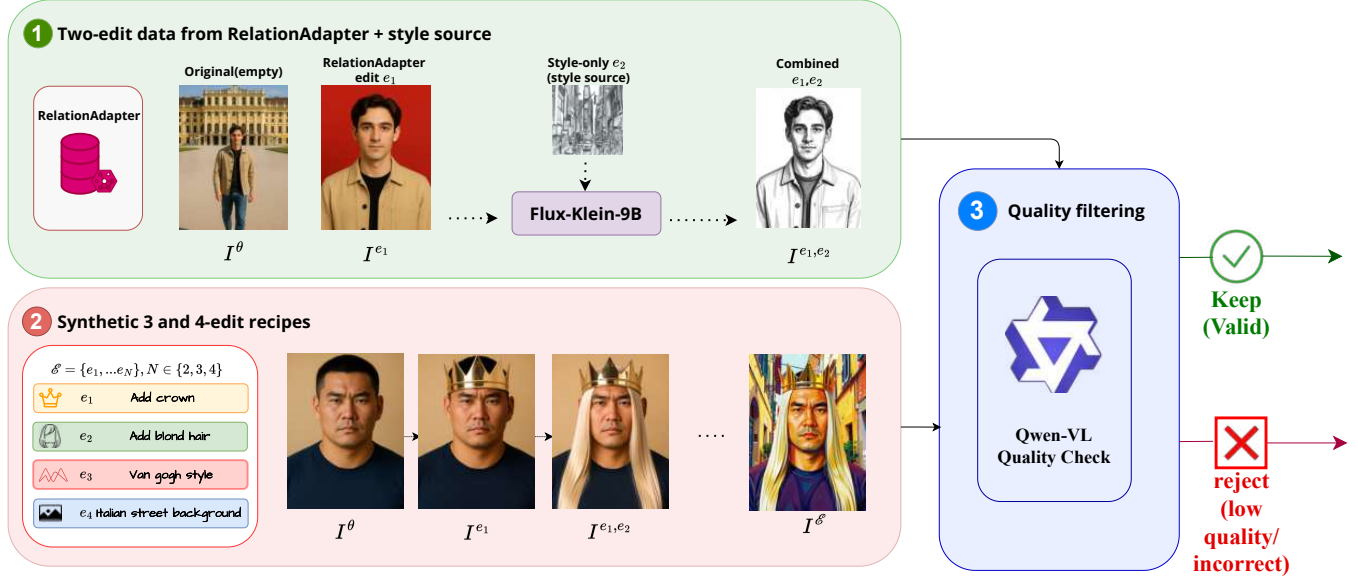


Fig. 7. **Overview of our multi-edit dataset generation pipeline.** For higher-order edit settings, we start from a base image and apply edits sequentially to obtain intermediate composites and the final full-edit image. For two-edit settings, we augment RelationAdapter pairs with an additional style transformation synthesized by FLUX.2-Klein. All generated samples are filtered with Qwen2.5-VL to verify that the intended edits are correctly expressed.

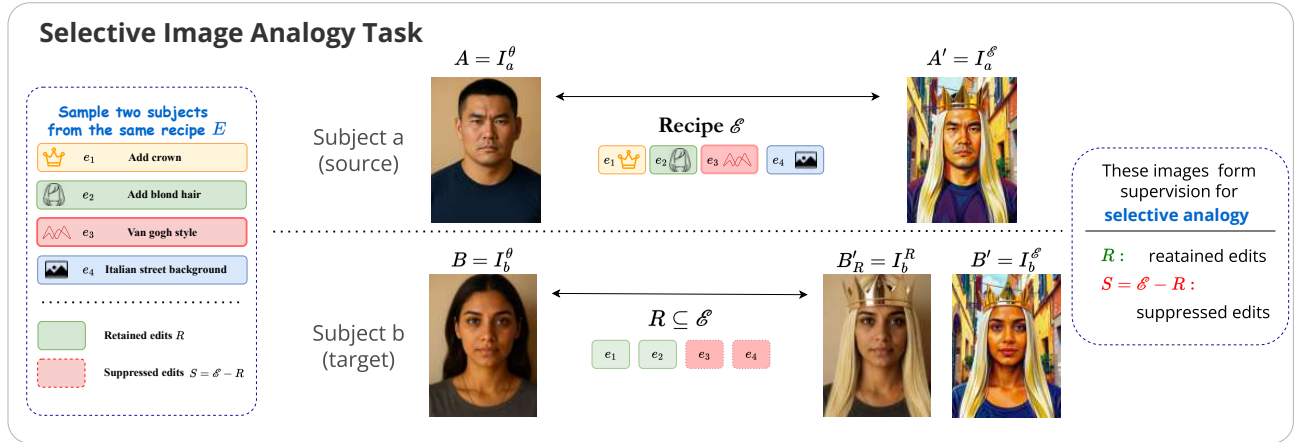


Fig. 8. **Per-sample storage layout.** Each leaf directory stores the source image, one reference image for each edit phrase, the sequence of composite images corresponding to retained subsets $\mathcal{R} \subseteq \mathcal{E}$, and a prompt .json file that maps every composite to a short natural-language description.

and for a four-edit recipe,

$$I^\emptyset \rightarrow I^{\{e_1\}} \rightarrow I^{\{e_1,e_2\}} \rightarrow I^{\{e_1,e_2,e_3\}} \rightarrow I^\mathcal{E}.$$

The three-edit group contains recipes such as one object addition with a style change and a background change, two object additions with a style change, or two object additions with a background change. The four-edit group contains a single recipe that combines two object additions, one background change, and one style change. The higher-order groups are less diverse than the two-edit subset, but they serve a different purpose: they teach the model to handle settings where several edits must be suppressed together. In

short, the two-edit data provides most of the edit diversity, while the three- and four-edit groups encourage the same suppression behavior in more complex multi-edit cases.

E.4 Vision-language filtering

The sequential synthesis pipeline can introduce artifacts. An intended edit may fail to appear in a composite, or an edit from an earlier step may be partially undone by a later step. To remove these failure cases, we filter every leaf directory with a vision-language

model. For each composite $I^{\mathcal{R}}$, we show the model the source image and the composite, then ask it to check two conditions: (i) every edit in $\mathcal{R}$ is present, and (ii) every edit in $\mathcal{E} \setminus \mathcal{R}$ is absent. Samples that fail either condition are discarded. We use Qwen2.5-VL-32B-Instruct for this filtering step, with separate text-only and image-conditioned passes when needed. The text-only pass is used to paraphrase the high-level transformation label, while the image-conditioned pass is used to verify style or appearance changes. The resulting dataset therefore provides not only the full transformation between A and A' , but also the intermediate partial composites needed for selective supervision. It is also filtered so that the desired edits are present and the excluded edits are absent.

E.5 Dataset Statistics and Composition Analysis

This section describes how the dataset is composed and how it is split into training and test sets. In total, it contains **1,845** samples, organized into **5** categories and **238** distinct edit-recipe *settings*. Each category is defined by two properties: the number of edits in the recipe, $N \in \{2, 3, 4\}$, and whether the subject is a stylized *creature* (the default, majority case) or a *non-creature* object such as a vehicle or prop. Across these categories, the dataset spans four fundamental edit types: object addition, style transfer, pose change, and facial expression change. Tab. 7 reports the full composition.

Train/test partitioning. Overall, **1,606** samples (87.05%) are used for training and **239** (12.95%) for testing, a ratio of roughly 6.72 : 1. This split is not applied uniformly across categories. Instead, the ratio in each category reflects the role that category plays in the dataset.

The 2edits creature category is by far the largest and most varied part of the dataset, with 1,478 samples across 210 settings. It supplies most of the visual and recipe diversity, so we use almost all of it for training (98.71%) and hold out only 19 samples for testing. Every other category is majority-test: 63.16% for 2edits non-creature, 58.67% for 3edits, 61.02% for 3edits non-creature, and 57.14% for 4edits. The reasoning is simple. The large two-edit creature pool provides the training signal, while the smaller and more narrowly built categories are reserved mainly for evaluation. They let us test whether the model’s selective-suppression behavior carries over to more complex recipes and to a different type of subject, rather than adding much training signal of their own.

The settings counts follow the same pattern. The 2edits category alone accounts for 210 of the 238 settings (88.3%), yet contributes only 19 test samples. The remaining four categories together contribute just 28 settings (11.7%), but send most of their samples to the test split.

Distribution across edit complexity. It is also useful to group samples purely by edit count, ignoring the creature and non-creature distinction. The subtotal rows of Tab. 7 give this view. Two-edit recipes dominate at 83.20% of all samples. This matches our design: the RelationAdapter-derived two-edit subset (Sec. E.2) supplies most of the visual and semantic diversity, while the three- and four-edit subsets (Sec. E.3) are deliberately smaller and built specifically to exercise multi-edit suppression, where two or more edits must be excluded at once.



Fig. 9. **Visual Ablation.** We show that when both ExLoRA and CSV are active, the retained edit is preserved the best, showing the necessity of both components. Note that AnaLoRA is always active as it is needed to capture all the provided edits.

As a result, the three-edit tier sends a much larger share of its samples to the test split (160/268, 59.7%) than the two-edit tier does (55/1,535, 3.6%), with the four-edit tier close behind at 24/42 (57.1%). The large two-edit pool teaches the model the underlying structure of edit interactions rather than letting it memorize specific edit combinations. The three- and four-edit samples then extend this to cases where several unwanted edits must be suppressed together. Together, the two give a balanced range of editing complexity, so the model generalizes across different numbers of edits while still suppressing edits reliably.

Approach for training with dataset. We build training instances by pairing samples. For each setting in the training split, we form every possible pair of samples within that setting, assigning one sample to the analogy source (A, A') and the other to the analogy target (B, B'), together with the edit descriptions for that setting.

Each pair then yields several training instances, one for each retained subset available in that category (Sec. E.1). A pair from a 2-edit setting yields 3 instances ($I^{\{e_1\}}, I^{\{e_2\}}, I^{\mathcal{E}}$), while a pair from a 2-edit non-creature setting yields 2 ($I^{\{e_1\}}, I^{\mathcal{E}}$). A pair from a 3-edit or 3-edit non-creature setting yields 3 ($I^{\{e_1\}}, I^{\{e_1, e_2\}}, I^{\mathcal{E}}$), and a pair from a 4-edit setting yields 4 ($I^{\{e_1\}}, I^{\{e_1, e_2\}}, I^{\{e_1, e_2, e_3\}}, I^{\mathcal{E}}$). Applied across all training settings, this produces 23,992 effective training instances, which we find sufficient for our selective image analogy setting. Tab. 8 summarizes this calculation.

F Visual Ablation

We qualitatively ablate the two selective suppression components of SIA in Fig. 9.

G Evaluation Metric Details

Selective image analogy places an unusual demand on evaluation: a good output must apply the retained edits, *avoid* the suppressed edit, and otherwise leave the query image B untouched. No single similarity score captures these three requirements at once. A

Table 7. **Dataset composition and train/test split.** Each category is defined by its edit count N and by whether the subject is a creature or a non-creature object. Settings is the number of distinct edit recipes in that category. Train % and Test % are shares within the category; % of Data is the category’s share of all 1,845 samples. Subtotal rows aggregate categories with the same N .

Category	N	Variant	Settings	Train	Test	Total	Train %	Test %	% of Data
2edits	2	Creature	210	1,459	19	1,478	98.71%	1.29%	80.11%
2edits non creature	2	Non-creature	3	21	36	57	36.84%	63.16%	3.09%
<i>Subtotal</i>	2	–	213	1,480	55	1,535	96.42%	3.58%	83.20%
3edits	3	Creature	16	62	88	150	41.33%	58.67%	8.13%
3edits non creature	3	Non-creature	4	46	72	118	38.98%	61.02%	6.40%
<i>Subtotal</i>	3	–	20	108	160	268	40.30%	59.70%	14.53%
4edits	4	Creature	5	18	24	42	42.86%	57.14%	2.28%
Total	–	–	238	1,606	239	1,845	87.05%	12.95%	100.00%

Table 8. **Same-setting pairing calculation for training-instance construction.** For each category, we sum $\binom{M}{2}$ over all training settings, where M is the number of training samples in a given setting, then multiply by the per-category edit multiplier to obtain the effective training instances. Per-setting sample counts M : 2-edit non-creature = 7, 8, 6; 3-edit, avg. $M \approx 3.9$; 3-edit non-creature = 10, 12, 13, 11; 4-edit = 3, 2, 4, 4, 5.

Category	Settings	Num. pairs	Multiplier	Instances
2-edit	210	7,572	3	22,716
2-edit non-creature	3	64	2	128
3-edit	16	104	3	312
3-edit non-creature	4	244	3	732
4-edit	5	26	4	104
Total	238	–	–	23,992

method that copies the full demonstration scores well on “*did the edits appear?*”, but fails on suppression. A method that barely changes B scores well on preservation but fails to apply anything. We therefore evaluate each output with a structured, *per-edit* protocol rather than a single holistic judgement.

The main paper reports five metrics: three VLM-judge metrics (*Retained Accuracy*, *Preservation*, and *Edit Recall*) and two CLIP-based metrics (*CLIP-Dir Image* and *CLIP-Dir Retained*); we additionally report a *Suppression* score in the ablation (Tab. 2). Below we give the full protocol behind these numbers: the structured VLM evaluation, the deterministic per-edit mapping, the consistency correction, and the SIA-aware extension of the LoRWeB [Manor et al. 2026] CLIP-directional metric. All metrics are computed for each (sample, method) pair and then aggregated in two stages: (i) the mean within each task triple (setting, data id, retained subset), and (ii) the mean across tasks. This two-step aggregation follows LoRWeB and prevents recipes with many retained-subset choices from dominating the final average.

G.1 VLM single-image protocol

Rather than asking the judge for a single overall rating, which tends to be noisy and hard to reproduce, we ask it to reason about *one edit at a time* and commit to a discrete label, which we then map to a

number through a fixed table. This keeps the judgement grounded in specific, checkable observations and removes most of the subjectivity from the numeric score.

For each result, we show the judge the four images $[A, A', B, \hat{B}']$ with the retained and suppressed edit lists. The judge follows a three-step evaluation protocol:

- **STEP 1.** Give a brief factual description of $\hat{B}'$. This anchors the later judgements in what is actually visible, rather than in what the edit lists say *should* be there.
- **STEP 2.** For each atomic edit $e \in \mathcal{E}$, describe the $A \rightarrow A'$ change in the aspect named by e , describe the corresponding $B \rightarrow \hat{B}'$ change, and assign a per-edit match label $m(e) \in \{a, b, c\}$. The label records how closely the output reproduces the *specific* demonstrated change, not merely whether the right category of edit occurred:
 - (a) $B \rightarrow \hat{B}'$ reproduces $A \rightarrow A'$ exactly, including the same specific instance.
 - (b) the same aspect changes, but with a different specific instance.
 - (c) the aspect is unchanged.

The judge also assigns an identity-match label $m_{\text{id}} \in \{a, b, c\}$ indicating whether $\hat{B}'$ preserves the identity, layout, and unrelated content of B .

- **STEP 3.** Map each per-edit match label deterministically to a binary detection flag $d(e) \in \{0, 1\}$ and an integer score $s(e) \in \{1, \dots, 10\}$ using Tab. 9. The identity-match label is mapped to a 1–10 Preservation score using the same score ranges.

The model returns a flat JSON object containing the description, the per-edit fields (m, d, s) for every $e \in \mathcal{E}$, and the identity-match label with its 1–10 Preservation score; the exact prompt and output schema are given in Sec. G.4. We extract these fields and apply the small consistency correction described below before aggregation.

Judges. We report results with two independent judges. The primary judge is Qwen3-VL-32B-Instruct [Bai et al. 2025], which is also used in the main paper. As a robustness check, we additionally report scores from Gemma 3 [Gemma Team et al. 2025]. The two

Table 9. Deterministic mapping from the per-edit match label in STEP 2 to the binary detection flag $d(e)$ and the 1–10 score $s(e)$ in STEP 3. The same score ranges are used to map the identity-match label to the Preservation score.

Role	Match $m(e)$	$d(e)$	$s(e)$ (1–10)
RETAIN	(a)	1	9–10
RETAIN	(b)	1	5–7
RETAIN	(c)	0	1
SUPPRESS	(a)	1	1
SUPPRESS	(b)	1	4–6
SUPPRESS	(c)	0	9–10

judges differ in absolute calibration, with Gemma 3 being more lenient overall, but they preserve the same relative ranking of methods.

Consistency correction. The 1–10 score and the binary detection flag are both derived from the same match label. However, VLM responses occasionally break this coupling, for example by assigning a high numeric score while also marking the edit as undetected. To keep Retained Accuracy consistent with the detection flags, we cap it at 2 when no retained edit is detected and at 5 when fewer than half of the retained edits are detected. No other corrections are applied to the metrics reported in the main paper.

G.2 VLM metric definitions

Let $s(e) \in \{1, \dots, 10\}$ denote the consistency-corrected per-edit score, and let $d(e) \in \{0, 1\}$ denote the detection flag. Recall that $\mathcal{R}$ is the retained-edit set and $\mathcal{S} = \mathcal{E} \setminus \mathcal{R}$ is the suppressed-edit set.

Retained Accuracy $\uparrow$. Retained Accuracy is the average per-edit score over the retained edits:

$$\text{RetAcc} = \frac{1}{|\mathcal{R}|} \sum_{e \in \mathcal{R}} s(e).$$

Suppression $\uparrow$. Suppression is the analogous average over the *suppressed* edits, and measures how well the output leaves out the edits the user chose to exclude:

$$\text{Supp} = \frac{1}{|\mathcal{S}|} \sum_{e \in \mathcal{S}} s(e).$$

Because the suppress role uses the inverted score ranges of Tab. 9, a high Suppression score means the suppressed edits are correctly *absent* from $\hat{B}'$. This metric targets the core requirement of selective transfer, which Preservation alone does not capture: a method that leaves B almost unchanged scores well on Preservation yet may still fail to remove a suppressed edit that the demonstration would otherwise introduce. Suppression is defined only for partial-transfer samples, where $\mathcal{S} \neq \emptyset$.

Preservation $\uparrow$. Preservation is the 1–10 score obtained from the STEP-2 identity-match label, using the RETAIN score ranges of Tab. 9: (a) $\Rightarrow$ 9–10, (b) $\Rightarrow$ 5–7, and (c) $\Rightarrow$ 1. It reflects whether identity, layout, and unrelated content of B are left intact. A no-op method that returns $\hat{B}' \approx B$ will naturally score close to 10 on this metric, which

is why Preservation must always be read together with Retained Accuracy and Suppression rather than on its own.

Edit Recall $\uparrow$. Edit Recall is the fraction of retained edits detected by the VLM:

$$\text{EditRecall} = \frac{1}{|\mathcal{R}|} \sum_{e \in \mathcal{R}} \mathbf{1}[d(e) = 1].$$

The indicator is 1 when edit e is detected and 0 otherwise. Unlike Retained Accuracy, Recall only asks *whether* each wanted edit appeared at all, not whether it matches the demonstrated form. A method can therefore reach high Recall while still scoring lower on Retained Accuracy.

G.3 CLIP directional metrics

The VLM metrics above are semantic and judge-dependent. We complement them with two lightweight, judge-independent CLIP directional metrics. The original CLIP-directional metric used by LoRWeB [Manor et al. 2026] assumes that $\hat{B}'$ should reproduce the *full* transformation from A to A' . In selective image analogy, this assumption only holds for full-transfer samples. When the user suppresses an edit, $\hat{B}'$ should intentionally *disagree* with the full $A \rightarrow A'$ transformation along the suppressed-edit axis. As a result, the original metric can incorrectly penalize a correct selective output on partial-transfer samples. To address this, we also report a text-anchored directional metric built from the per-edit captions stored with each sample, following the text-driven CLIP-direction recipe popularized by StyleCLIP [Patashnik et al. 2021].

Notation. Let ϕ_I and ϕ_T denote the CLIP image and text encoders (ViT-L/14, OpenAI weights). For a single (sample, method) tuple, define the image-space directions

$$\Delta I = \phi_I(\hat{B}') - \phi_I(B), \quad \Delta I_A = \phi_I(A') - \phi_I(A),$$

and the retained-text direction

$$\Delta T_R = \phi_T(t_R) - \phi_T(t_{\text{src}}),$$

where

$$t_{\text{src}} = \text{“a photo”}, \quad t_R = \text{“a photo with (retained desc.)”}$$

is built from the per-edit captions of the retained set $\mathcal{R}$.

CLIP-Dir Image $\uparrow$. This is the original LoRWeB image-image directional metric, which checks whether the output moves through image space in the same direction as the demonstrated edit:

$$\text{CLIPDirImage} = \cos(\Delta I_A, \Delta I).$$

We compute it only on full-transfer samples ($\mathcal{R} = \mathcal{E}$). It is reported as NaN on partial-transfer rows because ΔI_A also includes the suppressed edits in those cases and would penalize a correct selective output.

CLIP-Dir Retained $\uparrow$. This is our SIA-aware text-anchored directional metric:

$$\text{CLIPDirRetained} = \cos(\Delta I, \Delta T_R).$$

It replaces the demonstrated image-space direction with the direction described by the retained-edit *text*, so it is defined for every sample, including partial transfers, and measures whether $\hat{B}'$ moves

toward the retained-edit description regardless of what the suppressed edits do. It therefore serves as the selective-transfer counterpart of CLIPDirImage when no full-transfer ground truth is available.

Both CLIP metrics are judge-independent and are computed once for the full benchmark.

G.4 VLM Evaluation Prompt

For reproducibility, we include the evaluation prompt. It restates the three-step protocol of Sec. G.1. The box fixes the exact output schema, which we parse to obtain the per-edit fields used by the metrics in Sec. G.2. The VLM receives the four images $[A, A', B, \hat{B}]$ together with the retained- and suppressed-edit lists.

VLM Evaluation Prompt

You are evaluating a selective image analogy result. You are given:

- (1) Image A : the original demonstration image.
- (2) Image A' : the edited version of Image A .
- (3) Image B : the query image to edit.
- (4) Output $\hat{B}$: the generated result.
- (5) Retained edits: edits that should appear in the output.
- (6) Suppressed edits: edits that should not appear in the output.

STEP 1. Describe the output $\hat{B}$ briefly.

STEP 2. For each atomic edit e_i in the demonstrated set, describe the $A \rightarrow A'$ change in the aspect named by e_i , describe the corresponding $B \rightarrow \hat{B}$ change, and commit to a match label in $\{a, b, c\}$:

- (a) same aspect, same specific instance.
- (b) same aspect, different specific instance.
- (c) aspect unchanged.

STEP 3. Use the match label to deterministically commit to a detection flag and a 1–10 score using the following role-dependent mapping. For a **RETAIN** edit:

- (a) gives detected = 1 and score 9–10;
- (b) gives detected = 1 and score 5–7;
- (c) gives detected = 0 and score 1.

For a **SUPPRESS** edit:

- (a) gives detected = 1 and score 1;
- (b) gives detected = 1 and score 4–6;
- (c) gives detected = 0 and score 9–10.

Use the same score ranges to map the identity-match label to the preservation score. Return your answer as a flat JSON object:

```
{
  "step1": "<brieif description of B'>",
  "per_edit": [
    {
      "edit": "<edit phrase>",
      "role": "retain" | "suppress",
      "a_to_a_prime": "<sentence>",
      "b_to_b_hat": "<sentence>",
      "match": "a" | "b" | "c",
      "detected": true | false,
      "score": <integer 1..10>
    },
    ...
  ],
  "identity_match": "a" | "b" | "c",
  "preservation": <integer 1..10> }
```